3 Discrete Mathematics: Mathematical Logic (78)

Syllabus: Propositional and first order logic.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	0	1	0	1	0	1	1	0	1	1	0
2 Marks Count	0	0	1	0	0	0	0	0	0	0	0
Total Marks	0	1	2	1	0	1	1	0	1	1	**0**

Welcome to the "Discrete Mathematics: Mathematical Logic" chapter of your GATE Computer Science exam preparation. This section is designed to provide a comprehensive, exam-focused overview of the fundamental concepts, formulas, and problem-solving techniques you'll need to master. Mathematical Logic is the bedrock of computer science, providing the formal tools to reason about computation, verify program correctness, design circuits, and understand artificial intelligence. For the GATE CS exam, this subject typically carries a weightage of 5-8 marks. Questions often involve determining the validity of arguments, checking logical equivalence, translating natural language statements into logical expressions, and identifying tautologies or contradictions. A strong grasp of these concepts is indispensable not just for direct questions but also for understanding the logical underpinnings of other topics like set theory, relations, and algorithms.

Topic-wise Key Concepts

Propositional Logic

Propositional Logic, also known as propositional calculus, is the study of propositions (declarative sentences that are either true or false, but not both) and their combinations using logical connectives. It forms the most basic level of logical reasoning, focusing on the truth values of simple statements and how they combine to determine the truth values of complex statements.

- **Definition and Core Idea:** A proposition is a statement that is unequivocally true or false. Propositional logic uses symbols to represent propositions and logical connectives to form compound propositions, allowing us to analyze their truth values systematically.
- **Important Formulas, Theorems, and Results:**
  1. **Logical Connectives:**
     - Negation (NOT): $\neg p$ (read as "not p")
     - Conjunction (AND): $p \wedge q$ (read as "p and q")
     - Disjunction (OR): $p \vee q$ (read as "p or q")
     - Implication (IF...THEN): $p \rightarrow q$ (read as "if p then q" or "p implies q")
     - Biconditional (IF AND ONLY IF): $p \leftrightarrow q$ (read as "p if and only if q")
     - Exclusive OR (XOR): $p \oplus q$ (read as "p xor q")
     - NAND: $p \uparrow q \equiv \neg(p \wedge q)$
     - NOR: $p \downarrow q \equiv \neg(p \vee q)$
  2. **Truth Tables:** Essential for defining connectives and evaluating compound propositions.
     - $\neg p$: T if p is F, F if p is T.
     - $p \wedge q$: T if both p and q are T, else F.
     - $p \vee q$: T if p is T or q is T (or both), else F.
     - $p \rightarrow q$: F only if p is T and q is F, else T.
     - $p \leftrightarrow q$: T if p and q have the same truth value, else F.
     - $p \oplus q$: T if p and q have different truth values, else F.
  3. **Tautology, Contradiction, Contingency:**
     - **Tautology:** A compound proposition that is always true, regardless of the truth values of its atomic propositions. Example: $p \vee \neg p$.
     - **Contradiction (or Absurdity):** A compound proposition that is always false. Example: $p \wedge \neg p$.
     - **Contingency:** A compound proposition that is neither a tautology nor a contradiction (i.e., its truth value depends on the truth values of its atomic propositions).
  4. **Logical Equivalence ($\equiv$):** Two propositions P and Q are logically equivalent if $P \leftrightarrow Q$ is a tautology. This means they always have the same truth value.
  5. **Laws of Logic (Logical Equivalences):**
     - **Identity Laws:**
       - $p \wedge T \equiv p$
       - $p \vee F \equiv p$

- **Domination Laws:**
  - $p \lor T \equiv T$
  - $p \land F \equiv F$
- **Idempotent Laws:**
  - $p \lor p \equiv p$
  - $p \land p \equiv p$
- **Double Negation Law:** $\neg(\neg p) \equiv p$
- **Commutative Laws:**
  - $p \lor q \equiv q \lor p$
  - $p \land q \equiv q \land p$
- **Associative Laws:**
  - $(p \lor q) \lor r \equiv p \lor (q \lor r)$
  - $(p \land q) \land r \equiv p \land (q \land r)$
- **Distributive Laws:**
  - $p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$
  - $p \lor (q \land r) \equiv (p \lor q) \land (p \lor r)$
- **De Morgan's Laws:**
  - $\neg(p \land q) \equiv \neg p \lor \neg q$
  - $\neg(p \lor q) \equiv \neg p \land \neg q$
- **Absorption Laws:**
  - $p \lor (p \land q) \equiv p$
  - $p \land (p \lor q) \equiv p$
- **Inverse Laws (Complement Laws):**
  - $p \lor \neg p \equiv T$
  - $p \land \neg p \equiv F$
- **Implication Equivalences:**
  - $p \to q \equiv \neg p \lor q$
  - $p \to q \equiv \neg q \to \neg p$ (Contrapositive)
  - $\neg(p \to q) \equiv p \land \neg q$
- **Biconditional Equivalences:**
  - $p \leftrightarrow q \equiv (p \to q) \land (q \to p)$
  - $p \leftrightarrow q \equiv (\neg p \lor q) \land (\neg q \lor p)$
  - $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
  - $p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$

6. **Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF):**
   - **Literal:** A propositional variable or its negation (e.g., $p, \neg q$).
   - **Clause:** A disjunction of literals (e.g., $p \lor \neg q \lor r$).
   - **CNF:** A conjunction of clauses (e.g., $(p \lor \neg q) \land (r \lor s)$).
   - **Minterm:** A conjunction of literals where each variable appears exactly once (e.g., $p \land \neg q \land r$).
   - **Maxterm:** A disjunction of literals where each variable appears exactly once (e.g., $p \lor \neg q \lor r$).
   - **DNF:** A disjunction of minterms (e.g., $(p \land \neg q) \lor (r \land s)$).
   - Any propositional formula can be converted into an equivalent CNF or DNF.

- **Key Properties and Identities:** All the laws of logic listed above are crucial identities for simplification and proving equivalence. Understanding the truth conditions for each connective is fundamental.
- **Common Pitfalls or Tricky Points:**
  - Confusing implication ($p \to q$) with causation or "if and only if". Remember $p \to q$ is false only when p is true and q is false.
  - Incorrectly applying De Morgan's Laws, especially with more complex expressions.
  - Misinterpreting the biconditional; it implies mutual implication.
  - Errors in constructing truth tables, particularly for propositions with many variables.
  - Assuming logical equivalence based on partial truth values.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Truth Tables:** The most direct method for small propositions to check tautology, contradiction, or equivalence.
  - **Logical Equivalences:** Use the laws of logic to simplify expressions or prove equivalence without truth tables. This is often faster for complex expressions.
  - **Substitution:** Replace sub-expressions with their known equivalents.
  - **Proof by Contradiction:** To show P is a tautology, assume $\neg P$ is true and derive a contradiction. To show $P \equiv Q$, assume $\neg(P \leftrightarrow Q)$ is true and derive a contradiction.
  - **Conversion to CNF/DNF:** Useful for standardization and for certain proof methods like resolution.

First Order Logic (Predicate Logic)

First Order Logic (FOL), also known as Predicate Logic, extends propositional logic by allowing us to express more complex statements about objects, their properties, and relationships. It introduces predicates, variables, and quantifiers, enabling reasoning about "all" or "some" elements within a domain.

- **Definition and Core Idea:** FOL allows us to break down propositions into predicates (properties or relations) and arguments (variables or constants). It introduces quantifiers ($\forall$ for "for all" and $\exists$ for "there exists") to make statements about collections of objects, providing a richer expressive power than propositional logic.
- **Important Formulas, Theorems, and Results:**
  1. **Components of FOL:**
     - **Predicates:** Represent properties or relations (e.g., $P(x)$ for "x is prime", $L(x, y)$ for "x loves y").
     - **Variables:** Placeholders for objects (e.g., x, y, z).
     - **Constants:** Specific objects (e.g., "Socrates", "5").
     - **Functions:** Map objects to other objects (e.g., $f(x)$ for "father of x").
     - **Quantifiers:**
       - **Universal Quantifier ($\forall$):** "For all", "for every", "for each". $\forall x P(x)$ means $P(x)$ is true for every x in the domain.
       - **Existential Quantifier ($\exists$):** "There exists", "for some", "for at least one". $\exists x P(x)$ means there is at least one x in the domain for which $P(x)$ is true.
  2. **Scope of Quantifiers, Free and Bound Variables:**
     - The **scope** of a quantifier is the part of the formula to which it applies.
     - A variable is **bound** if it is within the scope of a quantifier using that variable.
     - A variable is **free** if it is not bound by any quantifier.
     - A formula with no free variables is called a **sentence** or **closed formula** and has a definite truth value.
  3. **Quantifier Equivalences (Negation Rules):**
     - $\neg \forall x P(x) \equiv \exists x \neg P(x)$ (It is not true that all x have property P if and only if there exists an x that does not have property P).
     - $\neg \exists x P(x) \equiv \forall x \neg P(x)$ (It is not true that there exists an x with property P if and only if all x do not have property P).
  4. **Distributive Properties of Quantifiers:**
     - $\forall x(P(x) \land Q(x)) \equiv \forall x P(x) \land \forall x Q(x)$
     - $\exists x(P(x) \lor Q(x)) \equiv \exists x P(x) \lor \exists x Q(x)$
     - **Important Non-Equivalences:**
       - $\forall x(P(x) \lor Q(x))$ is NOT equivalent to $\forall x P(x) \lor \forall x Q(x)$. (e.g., "Every number is even or odd" is true, but "Every number is even or every number is odd" is false).
       - $\exists x(P(x) \land Q(x))$ is NOT equivalent to $\exists x P(x) \land \exists x Q(x)$. (e.g., "There is a number that is both even and prime" (2) is true, but "There is an even number and there is a prime number" is also true, but the existence of one doesn't imply the other is the same number).
  5. **Interchange of Quantifiers:**
     - $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
     - $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
     - **Order Matters for Mixed Quantifiers:**
       - $\forall x \exists y P(x, y)$ (For every x, there exists a y such that P(x,y)) is NOT equivalent to $\exists y \forall x P(x, y)$ (There exists a y such that for every x, P(x,y)).
       - Example: Let $P(x, y)$ be "x loves y". $\forall x \exists y L(x, y)$ means "Everyone loves someone". $\exists y \forall x L(x, y)$ means "There is someone whom everyone loves". These are clearly different.
  6. **Prenex Normal Form (PNF):** A formula is in PNF if all quantifiers are at the beginning of the formula, followed by a quantifier-free formula (called the matrix). Any FOL formula can be converted to an equivalent PNF.
- **Key Properties and Identities:** The quantifier negation rules are fundamental for manipulating and simplifying FOL expressions. Understanding the implications of quantifier order is critical.
- **Common Pitfalls or Tricky Points:**
  - **Incorrect Translation:** Translating natural language statements into FOL is a major source of errors. Pay close attention to words like "all", "some", "no", "only if", "unless".
    - "All A are B": $\forall x(A(x) \to B(x))$ (NOT $\forall x(A(x) \land B(x))$)
    - "Some A are B": $\exists x(A(x) \land B(x))$ (NOT $\exists x(A(x) \to B(x))$)
    - "No A are B": $\forall x(A(x) \to \neg B(x))$ or $\neg \exists x(A(x) \land B(x))$
  - **Scope Errors:** Misplacing parentheses or incorrectly determining the scope of a quantifier.
  - **Variable Binding:** Confusing free and bound variables, or using the same variable name for different purposes in nested scopes.

    - **Order of Mixed Quantifiers:** Assuming $\forall x \exists y P(x, y)$ is the same as $\exists y \forall x P(x, y)$.
- **Standard Problem-Solving Techniques or Shortcuts:**
    - **Step-by-step Translation:** Break down complex natural language sentences into smaller parts, translate each, and then combine them. Identify predicates, variables, and connectives first.
    - **Applying Quantifier Equivalences:** Use negation rules to simplify expressions or move negations inwards.
    - **Domain Awareness:** Always consider the domain of discourse when evaluating truth values or translating.
    - **Counterexamples:** To show a statement is false or not equivalent, try to find a specific example in a small domain that makes it false.

Logical Reasoning (Inference)

Logical Reasoning, or Inference, is the process of deriving conclusions from a set of premises using valid rules. It focuses on the structure of arguments to determine whether a conclusion necessarily follows from the premises, regardless of the actual truth of the premises.

- **Definition and Core Idea:** An argument is a sequence of propositions, where the first propositions are premises and the final proposition is the conclusion. An argument is **valid** if, whenever all its premises are true, the conclusion must also be true. The goal is to determine if an argument's structure guarantees the truth of its conclusion given true premises.
- **Important Formulas, Theorems, and Results:**
    1. **Argument Validity:** An argument with premises $P_1, P_2, \ldots, P_n$ and conclusion C is valid if the proposition $(P_1 \wedge P_2 \wedge \ldots \wedge P_n) \to C$ is a tautology.
    2. **Rules of Inference (for Propositional Logic):** These are fundamental valid argument forms.
        - **Modus Ponens (Law of Detachment):**

$$\begin{array}{l} p \to q \\ \underline{p \qquad} \\ \therefore q \end{array}$$

          Equivalent to $((p \to q) \wedge p) \to q$ being a tautology.
        - **Modus Tollens:**

$$\begin{array}{l} p \to q \\ \underline{\neg q \qquad} \\ \therefore \neg p \end{array}$$

          Equivalent to $((p \to q) \wedge \neg q) \to \neg p$ being a tautology.
        - **Hypothetical Syllogism:**

$$\begin{array}{l} p \to q \\ \underline{q \to r \quad} \\ \therefore p \to r \end{array}$$

          Equivalent to $((p \to q) \wedge (q \to r)) \to (p \to r)$ being a tautology.
        - **Disjunctive Syllogism:**

$$\begin{array}{l} p \vee q \\ \underline{\neg p \quad} \\ \therefore q \end{array}$$

          Equivalent to $((p \vee q) \wedge \neg p) \to q$ being a tautology.
        - **Addition:**

$$\begin{array}{l} \underline{p \qquad\quad} \\ \therefore p \vee q \end{array}$$

        - **Simplification:**

$$\frac{p \land q}{\therefore p}$$

- **Conjunction:**

$$\begin{array}{l} p \\ q \\ \hline \therefore p \land q \end{array}$$

- **Resolution:**

$$\begin{array}{l} p \lor q \\ \neg q \lor r \\ \hline \therefore p \lor r \end{array}$$

  This is particularly useful in automated theorem proving and for converting to CNF.
- **Constructive Dilemma:**

$$\begin{array}{l} (p \to q) \land (r \to s) \\ p \lor r \\ \hline \therefore q \lor s \end{array}$$

- **Destructive Dilemma:**

$$\begin{array}{l} (p \to q) \land (r \to s) \\ \neg q \lor \neg s \\ \hline \therefore \neg p \lor \neg r \end{array}$$

3. **Rules of Inference (for First Order Logic):**
   - **Universal Instantiation (UI):** If $\forall x P(x)$ is true, then $P(c)$ is true for any arbitrary individual c in the domain.

$$\frac{\forall x P(x)}{\therefore P(c)}$$

   - **Universal Generalization (UG):** If $P(c)$ is true for an arbitrary individual c in the domain, then $\forall x P(x)$ is true.

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

   - **Existential Instantiation (EI):** If $\exists x P(x)$ is true, then $P(c)$ is true for some specific individual c in the domain. This c must be a new constant not previously used in the proof.

$$\frac{\exists x P(x)}{\therefore P(c) \text{ (for some new c)}}$$

   - **Existential Generalization (EG):** If $P(c)$ is true for some specific individual c in the domain, then $\exists x P(x)$ is true.

$$\frac{P(c) \text{ for some specific } c}{\therefore \exists x P(x)}$$

4. **Proof Methods:**
   - **Direct Proof:** Assume premises are true, and use inference rules and logical equivalences to derive the conclusion.

        - **Indirect Proof (Proof by Contradiction):** To prove C from premises $P_1, \ldots, P_n$, assume $\neg C$ is true along with $P_1, \ldots, P_n$, and derive a contradiction (F).
        - **Proof by Contraposition:** To prove $p \to q$, prove its contrapositive $\neg q \to \neg p$.
- **Key Properties and Identities:** The validity of each inference rule is a key property. Understanding that these rules preserve truth is fundamental.
- **Common Pitfalls or Tricky Points:**
    - **Fallacies:** Invalid argument forms that resemble valid ones.
        - **Fallacy of Affirming the Consequent:** $((p \to q) \land q) \to p$ (NOT a tautology). Example: "If it rains, the ground is wet. The ground is wet. Therefore, it rained." (Could be wet from a sprinkler).
        - **Fallacy of Denying the Antecedent:** $((p \to q) \land \neg p) \to \neg q$ (NOT a tautology). Example: "If it rains, the ground is wet. It did not rain. Therefore, the ground is not wet." (Could be wet from a sprinkler).
    - **Confusing Validity with Soundness:** A valid argument can have false premises and a false conclusion. A **sound** argument is a valid argument with all true premises (and thus, a true conclusion). GATE questions usually focus on validity.
    - **Incorrect Application of Quantifier Rules:** Especially with EI and UG, ensure the constant chosen for instantiation is new (for EI) or truly arbitrary (for UG).
    - **Circular Reasoning:** Assuming the conclusion (or something equivalent to it) as a premise.
- **Standard Problem-Solving Techniques or Shortcuts:**
    - **Formal Proofs:** List premises and then systematically apply inference rules and logical equivalences to derive the conclusion, citing the rule for each step.
    - **Truth Tables (for small arguments):** Convert the argument into a single conditional statement $(P_1 \land \ldots \land P_n) \to C$ and check if it's a tautology.
    - **Resolution Principle:** Convert all premises and the negation of the conclusion into CNF clauses. Add the negation of the conclusion as a clause. If an empty clause ($\square$) can be derived through resolution, the original argument is valid.
    - **Look for Fallacies:** If an argument resembles a common fallacy, it's likely invalid.

Quick Formula Reference

Propositional Logic Equivalences

- $\neg(\neg p) \equiv p$ (Double Negation)
- $p \lor q \equiv q \lor p$ (Commutative)
- $p \land q \equiv q \land p$ (Commutative)
- $(p \lor q) \lor r \equiv p \lor (q \lor r)$ (Associative)
- $(p \land q) \land r \equiv p \land (q \land r)$ (Associative)
- $p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$ (Distributive)
- $p \lor (q \land r) \equiv (p \lor q) \land (p \lor r)$ (Distributive)
- $\neg(p \land q) \equiv \neg p \lor \neg q$ (De Morgan's)
- $\neg(p \lor q) \equiv \neg p \land \neg q$ (De Morgan's)
- $p \to q \equiv \neg p \lor q$
- $p \to q \equiv \neg q \to \neg p$ (Contrapositive)
- $\neg(p \to q) \equiv p \land \neg q$
- $p \leftrightarrow q \equiv (p \to q) \land (q \to p)$
- $p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$
- $p \lor \neg p \equiv T$ (Inverse Law)
- $p \land \neg p \equiv F$ (Inverse Law)
- $p \lor T \equiv T$ (Domination Law)
- $p \land F \equiv F$ (Domination Law)
- $p \land T \equiv p$ (Identity Law)
- $p \lor F \equiv p$ (Identity Law)
- $p \lor p \equiv p$ (Idempotent Law)
- $p \land p \equiv p$ (Idempotent Law)
- $p \lor (p \land q) \equiv p$ (Absorption Law)
- $p \land (p \lor q) \equiv p$ (Absorption Law)

Quantifier Equivalences

- $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- $\neg \exists x P(x) \equiv \forall x \neg P(x)$

- $\forall x(P(x) \land Q(x)) \equiv \forall x P(x) \land \forall x Q(x)$
- $\exists x(P(x) \lor Q(x)) \equiv \exists x P(x) \lor \exists x Q(x)$
- $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
- $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$

Rules of Inference (Propositional Logic)

- **Modus Ponens:** $(p \to q) \land p \implies q$
- **Modus Tollens:** $(p \to q) \land \neg q \implies \neg p$
- **Hypothetical Syllogism:** $(p \to q) \land (q \to r) \implies p \to r$
- **Disjunctive Syllogism:** $(p \lor q) \land \neg p \implies q$
- **Addition:** $p \implies p \lor q$
- **Simplification:** $p \land q \implies p$
- **Conjunction:** $p, q \implies p \land q$
- **Resolution:** $(p \lor q) \land (\neg q \lor r) \implies p \lor r$

Rules of Inference (First Order Logic)

- **Universal Instantiation (UI):** $\forall x P(x) \implies P(c)$
- **Universal Generalization (UG):** $P(c) \text{ for arbitrary } c \implies \forall x P(x)$
- **Existential Instantiation (EI):** $\exists x P(x) \implies P(c) \text{ (for new c)}$
- **Existential Generalization (EG):** $P(c) \text{ for specific } c \implies \exists x P(x)$

Important Tips for GATE

1. **Master Truth Tables:** They are the foundation. Be able to quickly construct and interpret truth tables for any compound proposition, especially for 2-3 variables. This helps in verifying tautologies, contradictions, and equivalences.
2. **Memorize Key Equivalences and Inference Rules:** Speed and accuracy in GATE questions heavily rely on your ability to recall and apply De Morgan's laws, implication equivalences, and the common rules of inference (Modus Ponens, Modus Tollens, etc.) without hesitation.
3. **Practice Natural Language Translation:** A significant portion of GATE questions involves translating English sentences into logical expressions (and vice-versa). Pay close attention to keywords like "all", "some", "no", "only if", "unless", and the correct use of quantifiers and connectives.
4. **Understand Quantifier Scope and Order:** Incorrectly interpreting the scope of quantifiers or swapping the order of mixed quantifiers ($\forall x \exists y$ vs. $\exists y \forall x$) is a very common mistake. Always visualize the domain and what each quantifier implies.
5. **Distinguish Validity from Soundness:** GATE questions usually ask about the *validity* of an argument (whether the conclusion logically follows from the premises), not its *soundness* (which also requires premises to be true in the real world). Focus on the logical structure.
6. **Systematic Approach to Proofs:** When asked to prove an argument's validity, don't guess. Start with the premises and systematically apply inference rules and logical equivalences, one step at a time, until you reach the conclusion. Clearly state the rule used for each step.
7. **Identify Common Fallacies:** Be aware of fallacies like "Affirming the Consequent" and "Denying the Antecedent". If an argument structure matches a known fallacy, you can quickly identify it as invalid.
8. **Time Management:** Some logical reasoning problems can be lengthy, especially those involving formal proofs or complex translations. If a question seems too time-consuming, make an educated guess or mark it for review if time permits, and move on to other questions.

3.1 First Order Logic (35)

3.1.1 First Order Logic: GATE CSE 1989 | Question: 14a

Symbolize the expression "Every mother loves her children" in predicate logic.

gate1989 descriptive first-order-logic mathematical-logic

Answer key

3.1.2 First Order Logic: GATE CSE 1991 | Question: 15,b

Consider the following first order formula:

$$\left(\begin{array}{c} \forall x \exists y : R(x,y) \\ \wedge \\ \forall x \forall y : (R(x,y) \implies \neg R(y,x)) \\ \wedge \\ \forall x \forall y \forall z : (R(x,y) \wedge R(y,z) \implies R(x,z)) \\ \wedge \\ \forall x : \neg R(x,x) \end{array}\right)$$

Does it have finite models?

Is it satisfiable? If so, give a countable model for it.

gate1991 mathematical-logic first-order-logic descriptive

Answer key

3.1.3 First Order Logic: GATE CSE 1992 | Question: 92,xv

Which of the following predicate calculus statements is/are valid?

A. $(\forall (x))P(x) \vee (\forall (x))Q(x) \implies (\forall (x))(P(x) \vee Q(x))$
B. $(\exists (x))P(x) \wedge (\exists (x))Q(x) \implies (\exists (x))(P(x) \wedge Q(x))$
C. $(\forall (x))(P(x) \vee Q(x)) \implies (\forall (x))P(x) \vee (\forall (x))Q(x)$
D. $(\exists (x))(P(x) \vee Q(x)) \implies \sim (\forall (x))P(x) \vee (\exists (x))Q(x)$

gate1992 mathematical-logic normal first-order-logic

Answer key

3.1.4 First Order Logic: GATE CSE 2003 | Question: 32

Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

A. $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$
B. $(\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta]$
C. $((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha]) \Rightarrow (\forall x)[\alpha]$
D. $(\forall x)[\alpha \Rightarrow \beta] \Rightarrow (((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

gatecse-2003 mathematical-logic first-order-logic normal

Answer key

3.1.5 First Order Logic: GATE CSE 2003 | Question: 33

Consider the following formula and its two interpretations I_1 and I_2.

$\alpha : (\forall x)\,[P_x \Leftrightarrow (\forall y)\,[Q_{xy} \Leftrightarrow \neg Q_{yy}]] \Rightarrow (\forall x)\,[\neg P_x]$

I_1 : Domain: the set of natural numbers

P_x = 'x is a prime number'

Q_{xy} = 'y divides x'

I_2 : same as I_1 except that P_x = 'x is a composite number'.

Which of the following statements is true?

A. I_1 satisfies α, I_2 does not
B. I_2 satisfies α, I_1 does not
C. Neither I_1 nor I_2 satisfies α
D. Both I_1 and I_2 satisfies α

gatecse-2003 mathematical-logic difficult first-order-logic

Answer key☟

3.1.6 First Order Logic: GATE CSE 2004 | Question: 23, ISRO2007-32

Identify the correct translation into logical notation of the following assertion.

Some boys in the class are taller than all the girls

Note: $\text{taller}(x, y)$ is true if x is taller than y.

A. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
B. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
C. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$
D. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$

gatecse-2004 mathematical-logic easy isro2007 first-order-logic

Answer key☟

3.1.7 First Order Logic: GATE CSE 2005 | Question: 41

What is the first order predicate calculus statement equivalent to the following?

"Every teacher is liked by some student"

A. $\forall(x)\ [\text{teacher}\ (x) \rightarrow \exists(y)\ [\text{student}\ (y) \rightarrow \text{likes}\ (y, x)]]$
B. $\forall(x)\ [\text{teacher}\ (x) \rightarrow \exists(y)\ [\text{student}\ (y) \wedge \text{likes}\ (y, x)]]$
C. $\exists(y)\forall(x)\ [\text{teacher}\ (x) \rightarrow [\text{student}\ (y) \wedge \text{likes}\ (y, x)]]$
D. $\forall(x)\ [\text{teacher}\ (x) \wedge \exists(y)\ [\text{student}\ (y) \rightarrow \text{likes}\ (y, x)]]$

gatecse-2005 mathematical-logic easy first-order-logic

Answer key☟

3.1.8 First Order Logic: GATE CSE 2006 | Question: 26

Which one of the first order predicate calculus statements given below correctly expresses the following English statement?

Tigers and lions attack if they are hungry or threatened.

A. $\forall x[(\text{tiger}(x) \wedge \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)]$
B. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \wedge \text{attacks}(x)]$
C. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \text{attacks}(x) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x))]$
D. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)]$

gatecse-2006 mathematical-logic normal first-order-logic tricky

Answer key

3.1.9 First Order Logic: GATE CSE 2007 | Question: 22

Let $\text{Graph}(x)$ be a predicate which denotes that x is a graph. Let $\text{Connected}(x)$ be a predicate which denotes that x is connected. Which of the following first order logic sentences **DOES NOT** represent the statement:

"Not every graph is connected"

A. $\neg\forall x \left(\text{Graph}(x) \implies \text{Connected}(x)\right)$

B. $\exists x \left(\text{Graph}(x) \wedge \neg \text{Connected}(x)\right)$

C. $\neg\forall x \left(\neg \text{Graph}(x) \vee \text{Connected}(x)\right)$

D. $\forall x \left(\text{Graph}(x) \implies \neg \text{Connected}(x)\right)$

gatecse-2007 mathematical-logic easy first-order-logic

Answer key

3.1.10 First Order Logic: GATE CSE 2008 | Question: 30

Let fsa and pda be two predicates such that $\text{fsa}(x)$ means x is a finite state automaton and $\text{pda}(y)$ means that y is a pushdown automaton. Let equivalent be another predicate such that $\text{equivalent}(a, b)$ means a and b are equivalent. Which of the following first order logic statements represent the following?

Each finite state automaton has an equivalent pushdown automaton

A. $(\forall x \text{ fsa}\,(x)) \implies (\exists y \text{ pda}\,(y) \wedge \text{equivalent}\,(x, y))$
B. $\neg\forall y\,(\exists x \text{ fsa}\,(x) \implies \text{pda}\,(y) \wedge \text{equivalent}\,(x, y))$
C. $\forall x \exists y\,(\text{fsa}\,(x) \wedge \text{pda}\,(y) \wedge \text{equivalent}\,(x, y))$
D. $\forall x \exists y\,(\text{fsa}\,(y) \wedge \text{pda}\,(x) \wedge \text{equivalent}\,(x, y))$

gatecse-2008 easy mathematical-logic first-order-logic

Answer key

3.1.11 First Order Logic: GATE CSE 2009 | Question: 23

Which one of the following is the most appropriate logical formula to represent the statement?

"Gold and silver ornaments are precious".

The following notations are used:

- $G(x)$: x is a gold ornament
- $S(x)$: x is a silver ornament
- $P(x)$: x is precious

A. $\forall x(P(x) \implies (G(x) \wedge S(x)))$

B. $\forall x((G(x) \wedge S(x)) \implies P(x))$

C. $\exists x((G(x) \wedge S(x)) \implies P(x))$

D. $\forall x((G(x) \vee S(x)) \implies P(x))$

gatecse-2009 mathematical-logic easy first-order-logic

Answer key

3.1.12 First Order Logic: GATE CSE 2009 | Question: 26

Consider the following well-formed formulae:

I. $\neg\forall x(P(x))$
II. $\neg\exists x(P(x))$
III. $\neg\exists x(\neg P(x))$
IV. $\exists x(\neg P(x))$

Which of the above are equivalent?

A. I and III

B. I and IV

C. II and III

D. II and IV

gatecse-2009 mathematical-logic normal first-order-logic

Answer key

3.1.13 First Order Logic: GATE CSE 2010 | Question: 30

Suppose the predicate $F(x, y, t)$ is used to represent the statement that person x can fool person y at time t.

Which one of the statements below expresses best the meaning of the formula,

$$\forall x \exists y \exists t (\neg F(x, y, t))$$

A. Everyone can fool some person at some time
B. No one can fool everyone all the time
C. Everyone cannot fool some person all the time
D. No one can fool some person at some time

gatecse-2010 mathematical-logic easy first-order-logic

Answer key

3.1.14 First Order Logic: GATE CSE 2011 | Question: 30

Which one of the following options is CORRECT given three positive integers x, y and z, and a predicate

$$P(x) = \neg(x = 1) \wedge \forall y (\exists z (x = y * z) \Rightarrow (y = x) \vee (y = 1))$$

A. $P(x)$ being true means that x is a prime number
B. $P(x)$ being true means that x is a number other than 1
C. $P(x)$ is always true irrespective of the value of x
D. $P(x)$ being true means that x has exactly two factors other than 1 and x

gatecse-2011 mathematical-logic normal first-order-logic

Answer key

3.1.15 First Order Logic: GATE CSE 2012 | Question: 13

What is the correct translation of the following statement into mathematical logic?

“Some real numbers are rational”

A. $\exists x(\text{real}(x) \vee \text{rational}(x))$
B. $\forall x(\text{real}(x) \rightarrow \text{rational}(x))$
C. $\exists x(\text{real}(x) \wedge \text{rational}(x))$
D. $\exists x(\text{rational}(x) \rightarrow \text{real}(x))$

gatecse-2012 mathematical-logic easy first-order-logic

Answer key

3.1.16 First Order Logic: GATE CSE 2013 | Question: 27

What is the logical translation of the following statement?

"None of my friends are perfect."

A. $\exists x(F(x) \wedge \neg P(x))$
B. $\exists x(\neg F(x) \wedge P(x))$
C. $\exists x(\neg F(x) \wedge \neg P(x))$
D. $\neg \exists x(F(x) \wedge P(x))$

gatecse-2013 mathematical-logic easy first-order-logic

Answer key

3.1.17 First Order Logic: GATE CSE 2013 | Question: 47

Which one of the following is **NOT** logically equivalent to $\neg \exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

A. $\forall x(\exists z(\neg \beta) \rightarrow \forall y(\alpha))$
B. $\forall x(\forall z(\beta) \rightarrow \exists y(\neg \alpha))$
C. $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg \beta))$
D. $\forall x(\exists y(\neg \alpha) \rightarrow \exists z(\neg \beta))$

mathematical-logic normal marks-to-all gatecse-2013 first-order-logic

Answer key

3.1.18 First Order Logic: GATE CSE 2014 | Set 1 | Question: 1

Consider the statement

"Not all that glitters is gold”

Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?

A. $\forall x : \text{glitters}(x) \Rightarrow \neg\text{gold}(x)$
B. $\forall x : \text{gold}(x) \Rightarrow \text{glitters}(x)$
C. $\exists x : \text{gold}(x) \wedge \neg\text{glitters}(x)$
D. $\exists x : \text{glitters}(x) \wedge \neg\text{gold}(x)$

gatecse-2014-set1 mathematical-logic first-order-logic

Answer key☟

3.1.19 First Order Logic: GATE CSE 2014 | Set 3 | Question: 53

The CORRECT formula for the sentence, "not all Rainy days are Cold" is

A. $\forall d(\text{Rainy}(d) \wedge \sim\text{Cold}(d))$
B. $\forall d(\sim\text{Rainy}(d) \to \text{Cold}(d))$
C. $\exists d(\sim\text{Rainy}(d) \to \text{Cold}(d))$
D. $\exists d(\text{Rainy}(d) \wedge \sim\text{Cold}(d))$

gatecse-2014-set3 mathematical-logic easy first-order-logic

Answer key☟

3.1.20 First Order Logic: GATE CSE 2015 | Set 2 | Question: 55

Which one of the following well-formed formulae is a tautology?

A. $\forall x\, \exists y\, R(x,y) \leftrightarrow \exists y\, \forall x\, R(x,y)$
B. $(\forall x\, [\exists y\, R(x,y) \to S(x,y)]) \to \forall x\, \exists y\, S(x,y)$
C. $[\forall x\, \exists y\, (P(x,y) \to R(x,y))] \leftrightarrow [\forall x\, \exists y\, (\neg P(x,y) \vee R(x,y))]$
D. $\forall x\, \forall y\, P(x,y) \to \forall x\, \forall y\, P(y,x)$

gatecse-2015-set2 mathematical-logic normal first-order-logic

Answer key☟

3.1.21 First Order Logic: GATE CSE 2016 | Set 2 | Question: 27

Which one of the following well-formed formulae in predicate calculus is **NOT** valid ?

A. $(\forall_x p(x) \implies \forall_x q(x)) \implies (\exists_x \neg p(x) \vee \forall_x q(x))$
B. $(\exists x p(x) \vee \exists x q(x)) \implies \exists x (p(x) \vee q(x))$
C. $\exists x (p(x) \wedge q(x)) \implies (\exists x p(x) \wedge \exists x q(x))$
D. $\forall x (p(x) \vee q(x)) \implies (\forall x p(x) \vee \forall x q(x))$

gatecse-2016-set2 mathematical-logic first-order-logic normal

Answer key☟

3.1.22 First Order Logic: GATE CSE 2017 | Set 1 | Question: 02

Consider the first-order logic sentence $F : \forall x(\exists y R(x,y))$. Assuming non-empty logical domains, which of the sentences below are *implied* by F?

I. $\exists y(\exists x R(x,y))$
II. $\exists y(\forall x R(x,y))$
III. $\forall y(\exists x R(x,y))$
IV. $\neg\exists x(\forall y \neg R(x,y))$

A. IV only   B. I and IV only   C. II only   D. II and III only

gatecse-2017-set1 mathematical-logic first-order-logic

Answer key☟

3.1.23 First Order Logic: GATE CSE 2018 | Question: 28

Consider the first-order logic sentence

$$\varphi \equiv \exists\, s \,\exists\, t \,\exists\, u \,\forall\, v \,\forall\, w \forall\, x \,\forall\, y \,\psi(s,t,u,v,w,x,y)$$

where $\psi(s,t,u,v,w,x,y,)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

A. There exists at least one model of φ with universe of size less than or equal to 3
B. There exists no model of φ with universe of size less than or equal to 3
C. There exists no model of φ with universe size of greater than 7
D. Every model of φ has a universe of size equal to 7

gatecse-2018 mathematical-logic normal first-order-logic two-marks

Answer key☟

3.1.24 First Order Logic: GATE CSE 2019 | Question: 35

Consider the first order predicate formula φ:

$\forall x[(\forall z\, z|x \Rightarrow ((z=x) \vee (z=1))) \rightarrow \exists w(w>x) \wedge (\forall z\, z|w \Rightarrow ((w=z) \vee (z=1)))]$

Here $a \mid b$ denotes that $'a \text{ divides } b'$, where a and b are integers. Consider the following sets:

- S_1 : $\{1,2,3,\dots,100\}$
- S_2 : Set of all positive integers
- S_3 : Set of all integers

Which of the above sets satisfy φ?

A. S_1 and S_2 B. S_1 and S_3 C. S_2 and S_3 D. S_1, S_2 and S_3

gatecse-2019 discrete-mathematics mathematical-logic first-order-logic two-marks normal

Answer key☟

3.1.25 First Order Logic: GATE CSE 2020 | Question: 39

Which one of the following predicate formulae is NOT logically valid?

Note that W is a predicate formula without any free occurrence of x.

A. $\forall x(p(x) \vee W) \equiv \forall x\, p(x) \vee W$
B. $\exists x(p(x) \wedge W) \equiv \exists x\, p(x) \wedge W$
C. $\forall x(p(x) \rightarrow W) \equiv \forall x\, p(x) \rightarrow W$
D. $\exists x(p(x) \rightarrow W) \equiv \forall x\, p(x) \rightarrow W$

gatecse-2020 first-order-logic mathematical-logic two-marks

Answer key☟

3.1.26 First Order Logic: GATE CSE 2023 | Question: 16

Geetha has a conjecture about integers, which is of the form

$$\forall x(P(x) \Longrightarrow \exists y Q(x,y)),$$

where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would *imply* Geetha's conjecture?

A. $\exists x(P(x) \wedge \forall y Q(x, y))$
B. $\forall x \forall y Q(x, y)$
C. $\exists y \forall x(P(x) \Longrightarrow Q(x, y))$
D. $\exists x(P(x) \wedge \exists y Q(x, y))$

gatecse-2023 mathematical-logic first-order-logic multiple-selects one-mark difficult

Answer key

3.1.27 First Order Logic: GATE CSE 2025 | Set 1 | Question: 38

Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: "Everyone has exactly one mother"?

The meanings of the predicates used are:

- mother (y, x) : y is the mother of x
- noteq (x, y) : x and y are not equal

A. $\forall x \exists y \exists z(\text{mother}(y, x) \wedge \neg \text{mother}(z, x))$
B. $\forall x \exists y[\text{mother}(y, x) \wedge \forall z(\text{noteq}(z, y) \rightarrow \neg \text{mother}(z, x))]$
C. $\forall x \forall y[\text{mother}(y, x) \rightarrow \exists z(\text{mother}(z, x) \wedge \neg \text{note}\, q(z, y))]$
D. $\forall x \exists y[\text{mother}(y, x) \wedge \neg \exists z(\text{note}\, q(z, y) \wedge \text{mother}(z, x))]$

gatecse2025-set1 mathematical-logic first-order-logic multiple-selects two-marks

Answer key

3.1.28 First Order Logic: GATE CSE 2025 | Set 2 | Question: 5

Let $P(x)$ be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?

A. $\left(P(0) \wedge \left(\forall x[P(x) \Rightarrow P(x+1)]\right)\right) \Rightarrow \forall x P(x)\big)$
B. $\left(P(0) \wedge \left(\forall x[P(x) \Rightarrow P(x-1)]\right)\right) \Rightarrow \forall x P(x)\big)$
C. $\left(P(1000) \wedge \left(\forall x[P(x) \Rightarrow P(x-1)]\right)\right) \Rightarrow \forall x P(x)\big)$
D. $\left(P(1000) \wedge \left(\forall x[P(x) \Rightarrow P(x+1)]\right)\right) \Rightarrow \forall x P(x)\big)$

gatecse2025-set2 mathematical-logic first-order-logic one-mark

Answer key

3.1.29 First Order Logic: GATE CSE 2026 | Set 2 | Question: 1

For two different persons x and y, the predicate $M(x, y)$ denotes that x knows y. Consider the following statement.

There is a person who does not know anyone else, but that person is known by everyone else.

Which one of the following expressions represents the above statement?

A. $(\exists y)(\forall x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
B. $(\forall y)(\exists x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
C. $(\exists y)(\exists x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
D. $(\forall y)(\forall x)((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$

gatecse-2026-set2 mathematical-logic first-order-logic one-mark

Answer key

3.1.30 First Order Logic: GATE IT 2004 | Question: 3

Let $a(x,y), b(x,y,)$ and $c(x,y)$ be three statements with variables x and y chosen from some universe. Consider the following statement:

$$(\exists x)(\forall y)[(a(x,y) \wedge b(x,y)) \wedge \neg c(x,y)]$$

Which one of the following is its equivalent?

A. $(\forall x)(\exists y)[(a(x,y) \vee b(x,y)) \rightarrow c(x,y)]$
B. $(\exists x)(\forall y)[(a(x,y) \vee b(x,y)) \wedge \neg c(x,y)]$
C. $\neg(\forall x)(\exists y)[(a(x,y) \wedge b(x,y)) \rightarrow c(x,y)]$
D. $\neg(\forall x)(\exists y)[(a(x,y) \vee b(x,y)) \rightarrow c(x,y)]$

gateit-2004 mathematical-logic normal discrete-mathematics first-order-logic

Answer key

3.1.31 First Order Logic: GATE IT 2005 | Question: 36

Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always **TRUE**?

A. $((\forall x\left(P\left(x\right) \vee Q\left(x\right)\right))) \implies ((\forall xP\left(x\right)) \vee (\forall xQ\left(x\right)))$
B. $(\forall x\left(P\left(x\right) \implies Q\left(x\right)\right)) \implies ((\forall xP\left(x\right)) \implies (\forall xQ\left(x\right)))$
C. $(\forall x\left(P\left(x\right)\right) \implies \forall x\left(Q\left(x\right)\right)) \implies (\forall x\left(P\left(x\right) \implies Q\left(x\right)\right))$
D. $(\forall x\left(P\left(x\right)\right) \Leftrightarrow (\forall x\left(Q\left(x\right)\right))) \implies (\forall x\left(P\left(x\right) \Leftrightarrow Q\left(x\right)\right))$

gateit-2005 mathematical-logic first-order-logic normal

Answer key

3.1.32 First Order Logic: GATE IT 2006 | Question: 21

Consider the following first order logic formula in which R is a binary relation symbol.

$$\forall x \forall y (R(x,y) \implies R(y,x))$$

The formula is

A. satisfiable and valid
B. satisfiable and so is its negation
C. unsatisfiable but its negation is valid
D. satisfiable but its negation is unsatisfiable

gateit-2006 mathematical-logic normal first-order-logic

Answer key

3.1.33 First Order Logic: GATE IT 2007 | Question: 21

Which one of these first-order logic formulae is valid?

A. $\forall x\left(P\left(x\right) \implies Q\left(x\right)\right) \implies (\forall xP\left(x\right) \implies \forall xQ\left(x\right))$
B. $\exists x\left(P\left(x\right) \vee Q\left(x\right)\right) \implies (\exists xP\left(x\right) \implies \exists xQ\left(x\right))$
C. $\exists x\left(P\left(x\right) \wedge Q\left(x\right)\right) \iff (\exists xP\left(x\right) \wedge \exists xQ\left(x\right))$
D. $\forall x \exists yP\left(x,y\right) \implies \exists y \forall xP\left(x,y\right)$

gateit-2007 mathematical-logic normal first-order-logic

Answer key

3.1.34 First Order Logic: GATE IT 2008 | Question: 21

Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first order formula with x as a free variable, and β is a first order formula with no free variable.

A. $[\beta \to (\exists x, \alpha(x))] \to [\forall x, \beta \to \alpha(x)]$
B. $[\exists x, \beta \to \alpha(x)] \to [\beta \to (\forall x, \alpha(x))]$
C. $[(\exists x, \alpha(x)) \to \beta] \to [\forall x, \alpha(x) \to \beta]$
D. $[(\forall x, \alpha(x)) \to \beta] \to [\forall x, \alpha(x) \to \beta]$

gateit-2008 first-order-logic normal

Answer key

3.1.35 First Order Logic: GATE IT 2008 | Question: 22

Which of the following is the negation of $[\forall x, \alpha \to (\exists y, \beta \to (\forall u, \exists v, y))]$

A. $[\exists x, \alpha \to (\forall y, \beta \to (\exists u, \forall v, y))]$
B. $[\exists x, \alpha \to (\forall y, \beta \to (\exists u, \forall v, \neg y))]$
C. $[\forall x, \neg\alpha \to (\exists y, \neg\beta \to (\forall u, \exists v, \neg y))]$
D. $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg y))]$

gateit-2008 mathematical-logic normal first-order-logic

Answer key

3.2 Logical Reasoning (3)

3.2.1 Logical Reasoning: GATE CSE 2012 | Question: 1

Consider the following logical inferences.

I_1: If it rains then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.

I_2: If it rains then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.

Which of the following is **TRUE**?

A. Both I_1 and I_2 are correct inferences
B. I_1 is correct but I_2 is not a correct inference
C. I_1 is not correct but I_2 is a correct inference
D. Both I_1 and I_2 are not correct inferences

gatecse-2012 mathematical-logic easy logical-reasoning

Answer key

3.2.2 Logical Reasoning: GATE CSE 2015 | Set 2 | Question: 3

Consider the following two statements.

- S_1: If a candidate is known to be corrupt, then he will not be elected
- S_2: If a candidate is kind, he will be elected

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

A. If a person is known to be corrupt, he is kind
B. If a person is not known to be corrupt, he is not kind
C. If a person is kind, he is not known to be corrupt
D. If a person is not kind, he is not known to be corrupt

gatecse-2015-set2 mathematical-logic normal logical-reasoning

Answer key

3.2.3 Logical Reasoning: GATE CSE 2015 | Set 3 | Question: 24

In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking the person replies the following

"The result of the toss is head if and only if I am telling the truth"

Which of the following options is correct?

A. The result is head
B. The result is tail
C. If the person is of Type 2, then the result is tail
D. If the person is of Type 1, then the result is tail

gatecse-2015-set3 mathematical-logic difficult logical-reasoning

Answer key

3.3 Propositional Logic (40)

3.3.1 Propositional Logic: GATE CSE 1987 | Question: 10e

Show that the conclusion $(r \to q)$ follows from the premises: $p, (p \to q) \vee (p \wedge (r \to q))$

gate1987 mathematical-logic propositional-logic proof descriptive

Answer key

3.3.2 Propositional Logic: GATE CSE 1988 | Question: 2vii

Define the validity of a well-formed formula(wff)?

gate1988 descriptive mathematical-logic propositional-logic

Answer key

3.3.3 Propositional Logic: GATE CSE 1989 | Question: 3-v

Which of the following well-formed formulas are equivalent?

A. $P \to Q$
B. $\neg Q \to \neg P$
C. $\neg P \vee Q$
D. $\neg Q \to P$

gate1989 normal mathematical-logic propositional-logic multiple-selects

Answer key

3.3.4 Propositional Logic: GATE CSE 1990 | Question: 3-x

Indicate which of the following well-formed formulae are valid:

A. $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \Rightarrow (P \Rightarrow R)$
B. $(P \Rightarrow Q) \Rightarrow (\neg P \Rightarrow \neg Q)$
C. $(P \wedge (\neg P \vee \neg Q)) \Rightarrow Q$
D. $(P \Rightarrow R) \vee (Q \Rightarrow R) \Rightarrow ((P \vee Q) \Rightarrow R)$

gate1990 normal mathematical-logic propositional-logic multiple-selects

Answer key

3.3.5 Propositional Logic: GATE CSE 1991 | Question: 03,xii

If F_1, F_2 and F_3 are propositional formulae such that $F_1 \wedge F_2 \to F_3$ and $F_1 \wedge F_2 \to \sim F_3$ are both tautologies, then which of the following is true:

A. Both F_1 and F_2 are tautologies
B. The conjunction $F_1 \wedge F_2$ is not satisfiable
C. Neither is tautologous
D. Neither is satisfiable
E. None of the above

gate1991 mathematical-logic normal propositional-logic multiple-selects

Answer key☟

3.3.6 Propositional Logic: GATE CSE 1992 | Question: 02,xvi

Which of the following is/are a tautology?

A. $a \vee b \rightarrow b \wedge c$
B. $a \wedge b \rightarrow b \vee c$
C. $a \vee b \rightarrow (b \rightarrow c)$
D. $a \rightarrow b \rightarrow (b \rightarrow c)$

gate1992 mathematical-logic easy propositional-logic

Answer key☟

3.3.7 Propositional Logic: GATE CSE 1992 | Question: 15.a

Use Modus ponens $(A, A \rightarrow B| = B)$ or resolution to show that the following set is inconsistent:

1. $Q(x) \rightarrow P(x) \vee \sim R(a)$
2. $R(a) \vee \sim Q(a)$
3. $Q(a)$
4. $\sim P(y)$

where x and y are universally quantified variables, a is a constant and P, Q, R are monadic predicates.

gate1992 normal mathematical-logic propositional-logic descriptive

Answer key☟

3.3.8 Propositional Logic: GATE CSE 1993 | Question: 18

Show that proposition C is a logical consequence of the formula

$$A \wedge (A \rightarrow (B \vee C)) \wedge (B \rightarrow \neg A)$$

using truth tables.

gate1993 mathematical-logic normal propositional-logic proof descriptive

Answer key☟

3.3.9 Propositional Logic: GATE CSE 1993 | Question: 8.2

The proposition $p \wedge (\sim p \vee q)$ is:

A. a tautology
B. logically equivalent to $p \wedge q$
C. logically equivalent to $p \vee q$
D. a contradiction
E. none of the above

gate1993 mathematical-logic easy propositional-logic

Answer key☟

3.3.10 Propositional Logic: GATE CSE 1994 | Question: 3.13

Let p and q be propositions. Using only the Truth Table, decide whether

- $p \iff q$ does not imply $p \rightarrow \neg q$

is **True** or **False**.

gate1994 mathematical-logic normal propositional-logic true-false

Answer key☟

3.3.11 Propositional Logic: GATE CSE 1995 | Question: 13

Obtain the principal (canonical) conjunctive normal form of the propositional formula

$$(p \wedge q) \vee (\neg q \wedge r)$$

where $\wedge$ is logical and, $\vee$ is inclusive or and $\neg$ is negation.

gate1995 mathematical-logic propositional-logic normal descriptive

Answer key☟

3.3.12 Propositional Logic: GATE CSE 1995 | Question: 2.19

If the proposition $\neg p \to q$ is true, then the truth value of the proposition $\neg p \vee (p \to q)$, where $\neg$ is negation, $\vee$ is inclusive OR and $\to$ is implication, is

A. True
B. Multiple Values
C. False
D. Cannot be determined

gate1995 mathematical-logic normal propositional-logic

Answer key☟

3.3.13 Propositional Logic: GATE CSE 1996 | Question: 2.3

Which of the following is NOT True?

(Read $\wedge$ as AND, $\vee$ as OR, $\neg$ as NOT, $\to$ as one way implication and $\leftrightarrow$ as two way implication)

A. $((x \to y) \wedge x) \to y$
B. $((\neg x \to y) \wedge (\neg x \to \neg y)) \to x$
C. $(x \to (x \vee y))$
D. $((x \vee y) \leftrightarrow (\neg x \to \neg y))$

gate1996 mathematical-logic normal propositional-logic

Answer key☟

3.3.14 Propositional Logic: GATE CSE 1997 | Question: 3.2

Which of the following propositions is a tautology?

A. $(p \vee q) \to p$
B. $p \vee (q \to p)$
C. $p \vee (p \to q)$
D. $p \to (p \to q)$

gate1997 mathematical-logic easy propositional-logic

Answer key☟

3.3.15 Propositional Logic: GATE CSE 1998 | Question: 1.5

What is the converse of the following assertion?

- I stay only if you go

A. I stay if you go
B. If I stay then you go
C. If you do not go then I do not stay
D. If I do not stay then you go

gate1998 mathematical-logic easy propositional-logic

Answer key☟

3.3.16 Propositional Logic: GATE CSE 1999 | Question: 14

Show that the formula $[(\sim p \vee q) \Rightarrow (q \Rightarrow p)]$ is not a tautology.

Let A be a tautology and B any other formula. Prove that $(A \vee B)$ is a tautology.

gate1999 mathematical-logic normal propositional-logic proof descriptive

Answer key☟

3.3.17 Propositional Logic: GATE CSE 2000 | Question: 2.7

Let a, b, c, d be propositions. Assume that the equivalence $a \Leftrightarrow (b \vee \neg b)$ and $b \Leftrightarrow c$ hold. Then the truth-value of the formula $(a \wedge b) \rightarrow (a \wedge c) \vee d$ is always

A. True
B. False
C. Same as the truth-value of b
D. Same as the truth-value of d

gatecse-2000 mathematical-logic normal propositional-logic

Answer key☟

3.3.18 Propositional Logic: GATE CSE 2001 | Question: 1.3

Consider two well-formed formulas in propositional logic

$$F_1 : P \Rightarrow \neg P \qquad F_2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$$

Which one of the following statements is correct?

A. F_1 is satisfiable, F_2 is valid
B. F_1 unsatisfiable, F_2 is satisfiable
C. F_1 is unsatisfiable, F_2 is valid
D. F_1 and F_2 are both satisfiable

gatecse-2001 mathematical-logic easy propositional-logic

Answer key☟

3.3.19 Propositional Logic: GATE CSE 2002 | Question: 1.8

"If X then Y unless Z" is represented by which of the following formulas in propositional logic? ("$\neg$" is negation, "$\wedge$" is conjunction, and "$\rightarrow$" is implication)

A. $(X \wedge \neg Z) \rightarrow Y$
B. $(X \wedge Y) \rightarrow \neg Z$
C. $X \rightarrow (Y \wedge \neg Z)$
D. $(X \rightarrow Y) \wedge \neg Z$

gatecse-2002 mathematical-logic normal propositional-logic

Answer key☟

3.3.20 Propositional Logic: GATE CSE 2002 | Question: 5b

Determine whether each of the following is a tautology, a contradiction, or neither ("$\vee$" is disjunction, "$\wedge$" is conjunction, "$\rightarrow$" is implication, "$\neg$" is negation, and "$\leftrightarrow$" is biconditional (if and only if).

1. $A \leftrightarrow (A \vee A)$
2. $(A \vee B) \rightarrow B$
3. $A \wedge (\neg(A \vee B))$

gatecse-2002 mathematical-logic easy descriptive propositional-logic

Answer key☟

3.3.21 Propositional Logic: GATE CSE 2003 | Question: 72

The following resolution rule is used in logic programming.
Derive clause $(P \vee Q)$ from clauses $(P \vee R), (Q \vee \neg R)$
Which of the following statements related to this rule is FALSE?

A. $((P \vee R) \wedge (Q \vee \neg R)) \Rightarrow (P \vee Q)$ is logically valid
B. $(P \vee Q) \Rightarrow ((P \vee R) \wedge (Q \vee \neg R))$ is logically valid
C. $(P \vee Q)$ is satisfiable if and only if $(P \vee R) \wedge (Q \vee \neg R)$ is satisfiable
D. $(P \vee Q) \Rightarrow \text{FALSE}$ if and only if both P and Q are unsatisfiable

gatecse-2003 mathematical-logic normal propositional-logic

Answer key☟

3.3.22 Propositional Logic: GATE CSE 2004 | Question: 70

The following propositional statement is $(P \implies (Q \vee R)) \implies ((P \wedge Q) \implies R)$

A. satisfiable but not valid
B. valid
C. a contradiction
D. None of the above

gatecse-2004 mathematical-logic normal propositional-logic

Answer key

3.3.23 Propositional Logic: GATE CSE 2005 | Question: 40

Let $P, Q,$ and R be three atomic propositional assertions. Let X denote $(P \vee Q) \rightarrow R$ and Y denote $(P \rightarrow R) \vee (Q \rightarrow R)$. Which one of the following is a tautology?

A. $X \equiv Y$
B. $X \rightarrow Y$
C. $Y \rightarrow X$
D. $\neg Y \rightarrow X$

gatecse-2005 mathematical-logic propositional-logic normal

Answer key

3.3.24 Propositional Logic: GATE CSE 2006 | Question: 27

Consider the following propositional statements:

- $P_1 : ((A \wedge B) \rightarrow C)) \equiv ((A \rightarrow C) \wedge (B \rightarrow C))$
- $P_2 : ((A \vee B) \rightarrow C)) \equiv ((A \rightarrow C) \vee (B \rightarrow C))$

Which one of the following is true?

A. P_1 is a tautology, but not P_2
B. P_2 is a tautology, but not P_1
C. P_1 and P_2 are both tautologies
D. Both P_1 and P_2 are not tautologies

gatecse-2006 mathematical-logic normal propositional-logic

Answer key

3.3.25 Propositional Logic: GATE CSE 2008 | Question: 31

P and Q are two propositions. Which of the following logical expressions are equivalent?

I. $P \vee \neg Q$
II. $\neg(\neg P \wedge Q)$
III. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$
IV. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

A. Only I and II
B. Only I, II and III
C. Only I, II and IV
D. All of I, II, III and IV

gatecse-2008 normal mathematical-logic propositional-logic

Answer key

3.3.26 Propositional Logic: GATE CSE 2009 | Question: 24

The binary operation $\Box$ is defined as follows

P	**Q**	**P$\Box$Q**
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to $P \vee Q$?

A. $\neg Q \Box \neg P$
B. $P \Box \neg Q$
C. $\neg P \Box Q$
D. $\neg P \Box \neg Q$

gatecse-2009 mathematical-logic easy propositional-logic

Answer key

3.3.27 Propositional Logic: GATE CSE 2014 | Set 1 | Question: 53

Which one of the following propositional logic formulas is TRUE when exactly two of p, q and r are TRUE?

A. $((p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
B. $(\sim (p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
C. $((p \rightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
D. $(\sim (p \leftrightarrow q) \wedge r) \wedge (p \wedge q \wedge \sim r)$

gatecse-2014-set1 mathematical-logic normal propositional-logic

Answer key

3.3.28 Propositional Logic: GATE CSE 2014 | Set 2 | Question: 53

Which one of the following Boolean expressions is NOT a tautology?

A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
B. $(a \rightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$
D. $a \rightarrow (b \rightarrow a)$

gatecse-2014-set2 mathematical-logic propositional-logic normal

Answer key

3.3.29 Propositional Logic: GATE CSE 2014 | Set 3 | Question: 1

Consider the following statements:

- P: Good mobile phones are not cheap
- Q: Cheap mobile phones are not good

L: P implies Q
M: Q implies P
N: P is equivalent to Q

Which one of the following about L, M, and N is CORRECT?

A. Only L is TRUE.
B. Only M is TRUE.
C. Only N is TRUE.
D. L, M and N are TRUE.

gatecse-2014-set3 mathematical-logic easy propositional-logic

Answer key

3.3.30 Propositional Logic: GATE CSE 2015 | Set 1 | Question: 14

Which one of the following is NOT equivalent to $p \leftrightarrow q$?

A. $(\neg p \vee q) \wedge (p \vee \neg q)$
B. $(\neg p \vee q) \wedge (q \rightarrow p)$
C. $(\neg p \wedge q) \vee (p \wedge \neg q)$
D. $(\neg p \wedge \neg q) \vee (p \wedge q)$

gatecse-2015-set1 mathematical-logic easy propositional-logic

Answer key

3.3.31 Propositional Logic: GATE CSE 2016 | Set 1 | Question: 1

Let p, q, r, s represents the following propositions.

- $p : x \in \{8, 9, 10, 11, 12\}$
- $q : x$ is a composite number.
- $r : x$ is a perfect square.
- $s : x$ is a prime number.

The integer $x \geq 2$ which satisfies $\neg\left((p \Rightarrow q) \wedge (\neg r \vee \neg s)\right)$ is ___________.

gatecse-2016-set1 mathematical-logic normal numerical-answers propositional-logic

Answer key

3.3.32 Propositional Logic: GATE CSE 2016 | Set 2 | Question: 01

Consider the following expressions:

i. $false$
ii. Q
iii. $true$
iv. $P \vee Q$
v. $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is ___________.

gatecse-2016-set2 mathematical-logic normal numerical-answers propositional-logic

Answer key

3.3.33 Propositional Logic: GATE CSE 2017 | Set 1 | Question: 01

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$
II. $q \Rightarrow p$
III. $(\neg q) \vee p$
IV. $(\neg p) \vee q$

A. I only
B. I and IV only
C. II only
D. II and III only

gatecse-2017-set1 mathematical-logic propositional-logic easy

Answer key

3.3.34 Propositional Logic: GATE CSE 2017 | Set 1 | Question: 29

Let p, q and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is

A. a tautology
B. a contradiction
C. always TRUE when p is FALSE
D. always TRUE when q is TRUE

gatecse-2017-set1 mathematical-logic propositional-logic

Answer key

3.3.35 Propositional Logic: GATE CSE 2017 | Set 2 | Question: 11

Let p, q, r denote the statements "*It is raining*", "*It is cold*", and "*It is pleasant*", respectively. Then the statement "*It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold*" is represented by

A. $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$
B. $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$
C. $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$
D. $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

gatecse-2017-set2 mathematical-logic propositional-logic

Answer key☟

3.3.36 Propositional Logic: GATE CSE 2021 | Set 1 | Question: 7

Let p and q be two propositions. Consider the following two formulae in propositional logic.

- $S_1 : (\neg p \wedge (p \vee q)) \rightarrow q$
- $S_2 : q \rightarrow (\neg p \wedge (p \vee q))$

Which one of the following choices is correct?

A. Both S_1 and S_2 are tautologies.
B. S_1 is a tautology but S_2 is not a tautology
C. S_1 is not a tautology but S_2 is a tautology
D. Neither S_1 nor S_2 is a tautology

gatecse-2021-set1 mathematical-logic propositional-logic one-mark

Answer key☟

3.3.37 Propositional Logic: GATE CSE 2021 | Set 2 | Question: 15

Choose the correct choice(s) regarding the following proportional logic assertion S:

$$S : ((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$$

A. S is neither a tautology nor a contradiction
B. S is a tautology
C. S is a contradiction
D. The antecedent of S is logically equivalent to the consequent of S

gatecse-2021-set2 multiple-selects mathematical-logic propositional-logic one-mark

Answer key☟

3.3.38 Propositional Logic: GATE CSE 2024 | Set 2 | Question: 2

Let p and q be the following propositions:

p : Fail grade can be given.
q : Student scores more than 50% marks.

Consider the statement: "*Fail grade cannot be given when student scores more than* 50% *marks.*"

Which one of the following is the CORRECT representation of the above statement in propositional logic?

A. $q \rightarrow \neg p$
B. $q \rightarrow p$
C. $p \rightarrow q$
D. $\neg p \rightarrow q$

gatecse-2024-set2 mathematical-logic propositional-logic one-mark

Answer key☟

3.3.39 Propositional Logic: GATE DS&AI 2024 | Question: 19

Let x and y be two propositions. Which of the following statements is a tautology /are tautologies?

A. $(\neg x \wedge y) \Rightarrow (y \Rightarrow x)$
B. $(x \wedge \neg y) \Rightarrow (\neg x \Rightarrow y)$
C. $(\neg x \wedge y) \Rightarrow (\neg x \Rightarrow y)$
D. $(x \wedge \neg y) \Rightarrow (y \Rightarrow x)$

gate-ds-ai-2024 mathematical-logic propositional-logic multiple-selects one-mark

Answer key☟

3.3.40 Propositional Logic: GATE IT 2004 | Question: 31

Let p, q, r and s be four primitive statements. Consider the following arguments:

- $P : [(\neg p \vee q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (\neg s \rightarrow q)$
- $Q : [(\neg p \wedge q) \wedge [q \rightarrow (p \rightarrow r)]] \rightarrow \neg r$
- $R : [[(q \wedge r) \rightarrow p] \wedge (\neg q \vee p)] \rightarrow r$
- $S : [p \wedge (p \rightarrow r) \wedge (q \vee \neg r)] \rightarrow q$

Which of the above arguments are valid?

A. P and Q only B. P and R only C. P and S only D. P, Q, R and S

gateit-2004 mathematical-logic normal propositional-logic

Answer key☟

Answer Keys

3.1.1	N/A	3.1.2	N/A	3.1.3	A	3.1.4	D	3.1.5	D
3.1.6	D	3.1.7	B	3.1.8	D	3.1.9	D	3.1.10	X
3.1.11	D	3.1.12	B	3.1.13	B	3.1.14	A	3.1.15	C
3.1.16	D	3.1.17	X	3.1.18	D	3.1.19	D	3.1.20	C
3.1.21	D	3.1.22	B	3.1.23	A	3.1.24	C	3.1.25	C
3.1.26	B;C	3.1.27	B;D	3.1.28	A	3.1.29	A	3.1.30	C
3.1.31	B	3.1.32	B	3.1.33	A	3.1.34	C	3.1.35	D
3.2.1	B	3.2.2	C	3.2.3	A	3.3.1	N/A	3.3.2	N/A
3.3.3	A;B;C	3.3.4	A	3.3.5	B	3.3.6	B	3.3.7	N/A
3.3.8	N/A	3.3.9	B	3.3.10	True	3.3.11	N/A	3.3.12	D
3.3.13	D	3.3.14	C	3.3.15	A	3.3.16	N/A	3.3.17	A
3.3.18	A	3.3.19	A	3.3.20	N/A	3.3.21	B	3.3.22	A
3.3.23	B	3.3.24	D	3.3.25	B	3.3.26	B	3.3.27	B
3.3.28	B	3.3.29	D	3.3.30	C	3.3.31	11	3.3.32	4
3.3.33	D	3.3.34	D	3.3.35	A	3.3.36	B	3.3.37	B;D
3.3.38	A	3.3.39	B;C;D	3.3.40	C				